Battery Interface Board

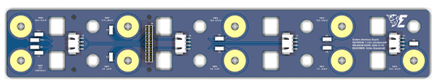
Variant: CHECKED

2026-11-10

Rev 1.2.0+ (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as ✗

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELASED - A board with this schematic has been sent to production.

DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

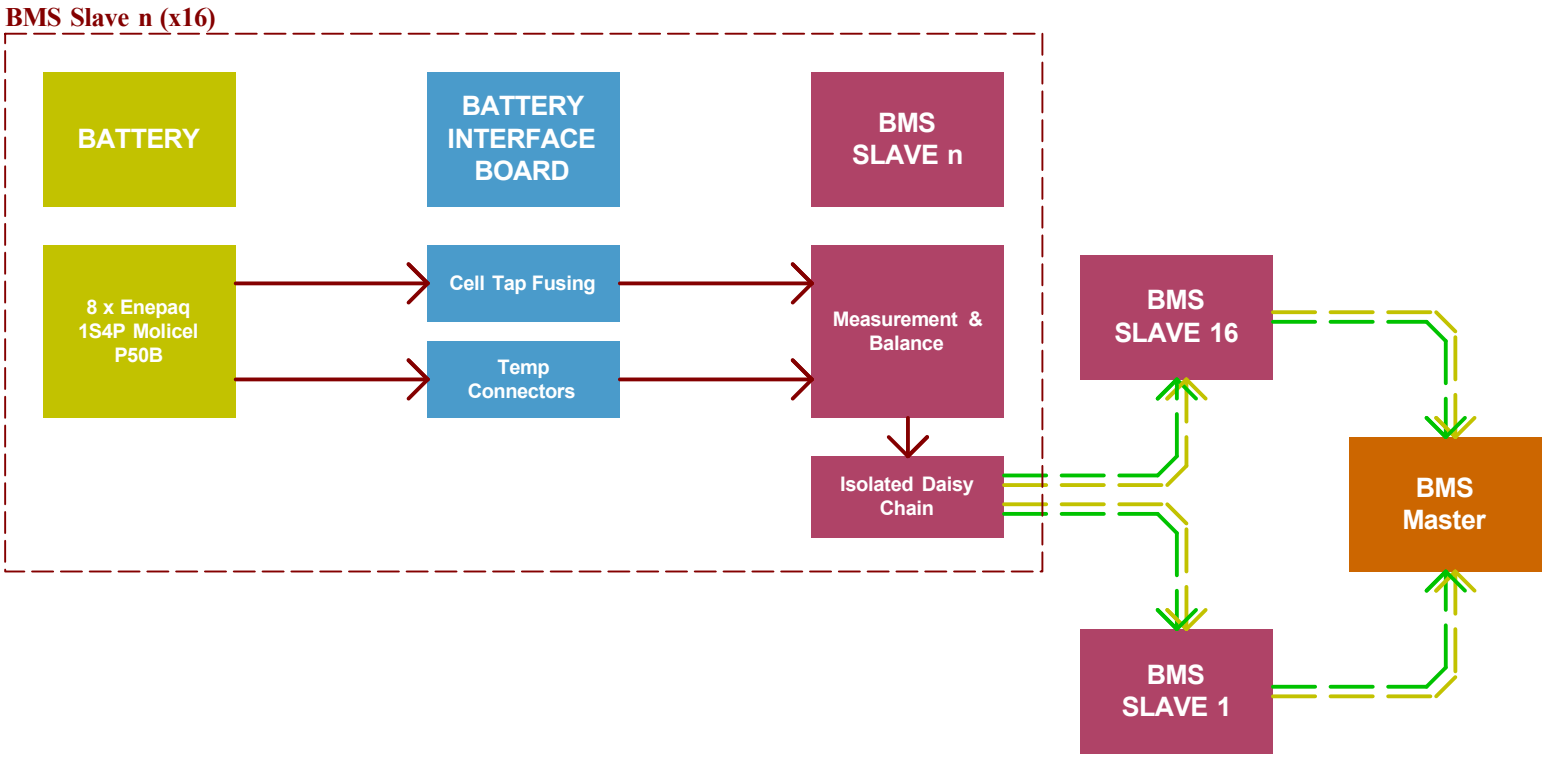
DESIGN NOTE: Example text for cautionary design notes.

DESIGN NOTE: Example text for critical design notes.

LAYOUT NOTE: Example text for critical layout guidelines.

Comments:	Company: Blue Devil Motor Sports		Variant: CHECKED	Git Hash: 0683f88
	Board Name: BATTERY_INTERFACE_BOARD		Project Name: Battery Interface Board	
Sheet Title:	File Name: battery_interface_board.kicad_sch	Designer: Aidan Brzezinski	Date: 2026-11-03	Revision: 1.2.0+ (Unreleased)
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 6

[2] Block Diagram



Target specifications:

Input voltage:	20 - 33.6 V
Tap current / protection	500 mA Fuse
Voltage Taps	9 x 2.5 - 4.2V
Temp Inputs	8 Enepaq, 1.30 - 2.44 V

Comments:	Company: **Blue Devil Motor Sports**		Variant: CHECKED	Git Hash: 0683f88
	Board Name: **BATTERY_INTERFACE_BOARD**		Project Name: **Battery Interface Board**	
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Aidan Brzezinski	Date: 2026-11-03	Revision: 1.2.0+ (Unreleased)
Sheet Path: /Block Diagram/		Reviewer:	Size: **A4**	Sheet: **2** of **6**

[3] Project Architecture

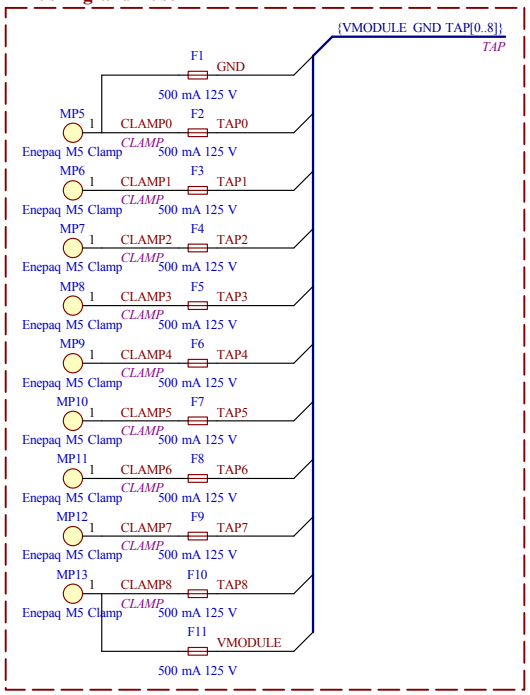
Connectors

Mechanical

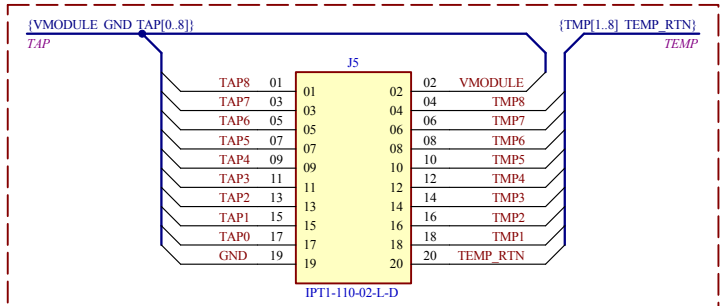
Comments:	Company: Blue Devil Motor Sports		Variant: CHECKED	Git Hash: 0683f88
	Board Name: **BATTERY_INTERFACE_BOARD**		Project Name: **Battery Interface Board**	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Aidan Brzezinski	Date: 2026-11-03	Revision: 1.2.0+ (Unreleased)
Sheet Path: /Project Architecture/		Reviewer:	Size: **A4**	Sheet: **3** of **6**

[4] Connectors

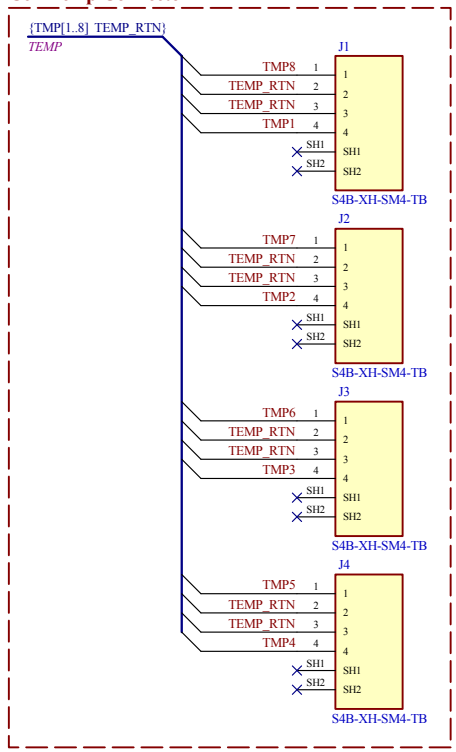
Bushing and Fuse



Board to Board Connector



Cell Temp Connector



Comments:	Company: **Blue Devil Motor Sports**		Variant: CHECKED	Git Hash: 0683f88
	Board Name: **BATTERY_INTERFACE_BOARD**		Project Name: **Battery Interface Board**	
Sheet Title: Connectors	File Name: connectors.kicad_sch	Designer: Aidan Brzezinski	Date: 2026-11-03	Revision: 1.2.0+ (Unreleased)
Sheet Path: /Project Architecture/Connectors/		Reviewer:	Size: **A4**	Sheet: **4** of **6**

[5] Mechanical

MECHANICAL NOTE: SLAVE BOARD MOUNTING (per corner, 4 places)
1. Spacer: Harwin R30-1611400
   Nylon 6.6 hex, female-female, M3 x 14 mm, 5.5 mm A/F, UL94 V-2.
2. Screws (x2): Essentra 50M030050P006
   M3 x 6 mm nylon pan head, Phillips.
   One from the top through the slave, one from the bottom through the interface board. ~4.4 mm thread engagement each end.
3. Holes H1-H4: 3.2 mm
4. Interface board: holes ONLY over gaps between busbars (bottom screw head ~2 mm below the board).
5. Stack height 14.0 mm = Samtec mated height:
   Interface board: IPT1-110-02-L-D (male)
   Slave board: IPS1-110-01-L-D (female)
6. Fully isolated: no metal path between boards.
   Hand-tight only (~0.2-0.3 Nm). No liquid thread-locker on nylon.

Holes

H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole

Silkscreen

SAE Electric logo

BMD Logo

Comments:	Company: **Blue Devil Motor Sports**		Variant: CHECKED	Git Hash: 0683f88
	Board Name: **BATTERY_INTERFACE_BOARD**		Project Name: **Battery Interface Board**	
Sheet Title: Mechanical	File Name: mechanical.kicad_sch	Designer: Aidan Brzezinski	Date: 2026-11-03	Revision: 1.2.0+ (Unreleased)
Sheet Path: /Project Architecture/Mechanical/		Reviewer:	Size: **A4**	Sheet: **5** of **6**

[7] Revision History

Version 1.2.0 - 2026-11-06

Fixed

- Fab doc impedence table/stack.

Version 1.1.0 - 2026-11-03

Fixed

- Fixed hole alignment with BMS_slave.

Version 1.0.0 - 2026-11-03

Comments:	Company: Blue Devil Motor Sports		Variant: CHECKED	Git Hash: 0683f88	
	Board Name: BATTERY_INTERFACE_BOARD		Project Name: Battery Interface Board		
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Aidan Brzezinski	Date: 2026-11-03	Revision: 1.2.0+ (Unreleased)	
Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 7 of 6	